SIM-Based Risky and Risk-Free Portfolio Allocation

CHEME 5660
Cornell University
October 1, 2026
Jeffrey D. Varner

LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

- **Construct SIM portfolio inputs.** Build expected-growth and positive-semidefinite growth-rate covariance estimates from market, beta, alpha, and residual-risk estimates.
- **Solve constrained allocations.** Formulate risky-only and risky/risk-free minimum-variance problems while distinguishing lending, borrowing, and short-sale constraints.
- **Validate the resulting portfolio.** Compare SIM and empirical allocations out of sample without treating a fitted frontier as realized performance.

1 From a Growth Model to a Portfolio Decision

The single-index model replaces a dense collection of sample covariances with one market factor and asset-specific residual risks. That parsimony is useful only when it is carried into an optimization problem with compatible units, feasible trading constraints, and honest validation. A smooth fitted frontier is a model output, not evidence that the selected portfolio will perform well.

For m risky assets sampled over a fixed interval Δt, let $\boldsymbol{\alpha}_g, \beta \in \mathbb{R}^m$ be growth-SIM intercepts and market loadings, let $\bar{g}_M \in \mathbb{R}$ and $\sigma_{g,M}^2 \geq 0$ be the market growth-rate mean and variance, and let $D_g := \operatorname{diag}(\sigma_{g,\varepsilon,1}^2, \dots, \sigma_{g,\varepsilon,m}^2)$ contain nonnegative growth-residual variances. The growth-based SIM portfolio inputs are

$$\bar{\mathbf{g}} = \boldsymbol{\alpha}_g + \beta\bar{g}_M, \qquad \Sigma_g = \sigma_{g,M}^2\beta\beta^{\mathsf{T}} + D_g. \tag{1}$$

Every quantity in Eq. 1 uses growth-rate units: $\bar{\mathbf{g}}$ and $\boldsymbol{\alpha}_g$ have units time^{-1}, while Σ_g and D_g have units time^{-2}. For the associated log return $\mathbf{r} = (\Delta t)\mathbf{g}$, $\Sigma_r = (\Delta t)^2\Sigma_g$. The GBM covariance rate is a third object, $C = \Sigma_r/\Delta t = \Delta t\,\Sigma_g$; it should not be inserted into a growth-rate variance formula without relabeling the objective.

Proposition 1.1. The SIM covariance is positive semidefinite

Let $\beta \in \mathbb{R}^m$, $\sigma_{g,M}^2 \geq 0$, and let $D_g \in \mathbb{R}^{m\times m}$ be diagonal with nonnegative entries. Then $\Sigma_g := \sigma_{g,M}^2\beta\beta^{\mathsf{T}} + D_g$ is symmetric positive semidefinite because, for every $\mathbf{x} \in \mathbb{R}^m$,

$$\mathbf{x}^{\mathsf{T}}\Sigma_g\mathbf{x} = \sigma_{g,M}^2(\beta^{\mathsf{T}}\mathbf{x})^2 + \sum_{i=1}^{m}(D_g)_{ii}x_i^2 \geq 0. \tag{2}$$

It is positive definite if every diagonal entry of D_g is strictly positive.

Positive semidefiniteness makes the variance objective convex. It does not establish that the covariance estimate is accurate. Residual cross-correlations, changing betas, and an inadequate market proxy can all produce a valid matrix that is a poor description of future co-movement.

2 Risky-Only Allocation with SIM Moments

Let $\mathbf{w} \in \mathbb{R}^m$ denote risky-asset dollar weights and let $g_\star \in \mathbb{R}$ be a target growth rate under the same sampling convention. A long-only minimum-variance allocation solves

$$\begin{aligned} \min_{\mathbf{w}\in\mathbb{R}^m} \quad & \frac{1}{2}\mathbf{w}^\mathsf{T}\Sigma_g\mathbf{w} \\ \text{subject to} \quad & \mathbf{1}^\mathsf{T}\mathbf{w} = 1, \\ & \bar{\mathbf{g}}^\mathsf{T}\mathbf{w} \geq g_\star, \\ & \mathbf{w} \geq \mathbf{0}. \end{aligned} \tag{3}$$

The inequality form asks for at least the target. When expected growth rates contain estimation noise, the optimum may overshoot the target; report the achieved growth and solver status rather than assuming equality. Varying $g_\star$ traces the modeled constrained boundary. If short sales are permitted, remove or replace the lower bounds and state any leverage or gross-exposure limit.

The rank-one structure also gives a computational shortcut. If D_g is positive definite, the Sherman–Morrison identity yields

$$\Sigma_g^{-1} = D_g^{-1} - \frac{\sigma_{g,M}^2 D_g^{-1}\beta\beta^\mathsf{T}D_g^{-1}}{1+\sigma_{g,M}^2\beta^\mathsf{T}D_g^{-1}\beta}. \tag{4}$$

Equation (4) explains why a factor model can scale well, but a constrained quadratic-program solver remains the appropriate tool for Eq. 3.

> **COMPANION NOTEBOOK**
> **L6b: SIM Risky-Asset Minimum-Variance Allocation →**
> Construct SIM and empirical covariance inputs for the same ticker universe, solve their constrained frontiers, and compare allocations and realized test-period wealth.

3 Adding a Horizon-Matched Risk-Free Asset

Let $g_f \in \mathbb{R}$ be a known continuously compounded risk-free growth rate, let $w_f \in \mathbb{R}$ be its dollar weight, and let $\mathbf{x} \in \mathbb{R}^m$ contain risky-asset weights. Because a risk-free payoff has zero growth-rate variance under the model, only $\mathbf{x}$ enters portfolio variance. The target-growth problem is

$$\begin{aligned} \min_{\mathbf{x},w_f} \quad & \frac{1}{2}\mathbf{x}^\mathsf{T}\Sigma_g\mathbf{x} \\ \text{subject to} \quad & \mathbf{1}^\mathsf{T}\mathbf{x} + w_f = 1, \\ & \bar{\mathbf{g}}^\mathsf{T}\mathbf{x} + g_f w_f \geq g_\star. \end{aligned} \tag{5}$$

Long-only risky assets impose $\mathbf{x} \geq \mathbf{0}$. Risk-free lending imposes $w_f \geq 0$; allowing $w_f < 0$ represents borrowing at the same growth rate g_f, a strong idealization. Eliminating $w_f = 1 - \mathbf{1}^\mathsf{T}\mathbf{x}$ rewrites the growth constraint as

$$(\bar{\mathbf{g}} - g_f\mathbf{1})^\mathsf{T}\mathbf{x} \geq g_\star - g_f. \tag{6}$$

If lending only is allowed, $w_f \geq 0$ additionally requires $\mathbf{1}^\mathsf{T}\mathbf{x} \leq 1$. If borrowing is allowed without limit, that inequality disappears. These are different feasible sets and should never share an unlabeled "capital allocation line."

Remark 3.1. When a Treasury STRIP is risk-free

A zero-coupon Treasury STRIP delivers a specified nominal payment only when its maturity and the investor's horizon match and the security is held to maturity. Before maturity its market price changes with yields. Inflation changes real purchasing power, and taxes, liquidity, and sovereign-credit assumptions remain. In these notes, "risk-free" means a model input with a known nominal horizon payoff, not an asset free of every economic risk.

4 The Capital Allocation Geometry and Its Limits

Let T be a feasible risky portfolio with expected growth $\bar{g}_T > g_f$ and growth-rate standard deviation $\sigma_{g,T} > 0$. If $y \geq 0$ is the total risky weight, combining y in T with $1 - y$ in the risk-free asset gives

$$\bar{g}_p = g_f + y(\bar{g}_T - g_f), \qquad \sigma_{g,p} = y\sigma_{g,T}. \tag{7}$$

For unrestricted lending and borrowing at one rate, the efficient risky fund maximizes $(\bar{g}_T - g_f)/\sigma_{g,T}$ and Eq. 7 traces a ray. With lending but no borrowing, only $0 \leq y \leq 1$ is available on that straight segment; beyond T, the constrained efficient set generally returns to the risky-asset boundary. Position limits, different borrowing rates, and short-sale constraints can introduce additional kinks.

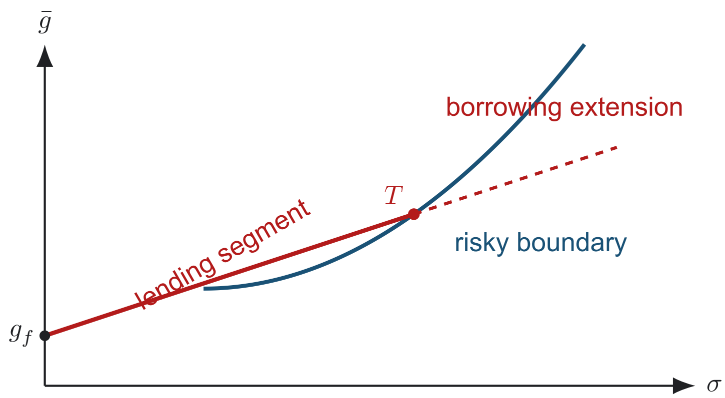

Figure 1: The solid red segment uses nonnegative risk-free weight: $0 \leq y \leq 1$. The dashed extension requires borrowing at the modeled risk-free rate. A constrained optimizer must enforce the regime actually available to the investor.

The classical separation theorem says investors with common inputs choose the same tangency fund and vary only y. This statement relies on mean–variance preferences, common beliefs, one frictionless borrowing/lending rate, and compatible unconstrained opportunities. Once constraints bind, different target growth rates can select different risky compositions; two-fund separation need not survive globally.

COMPANION NOTEBOOK

L6b: SIM Risky/Risk-Free Minimum-Variance Allocation →

Solve risky-only and risky/risk-free frontiers from SIM inputs, identify a candidate tangency allocation, and compare its test-period wealth with market and risk-free benchmarks.

COMPANION NOTEBOOK
L6b: Derivation of the SIM Covariance →
Expand the scalar covariance terms and identify exactly where market-orthogonality and diagonal-residual assumptions enter the result.

5 A Defensible Estimation and Validation Workflow

Use a training window to estimate $\boldsymbol{\alpha}_g$, β, D_g, $\bar{g}_M$, and $\sigma^2_{g,M}$. Freeze those inputs, solve the allocation, and then evaluate the fixed rule on later observations. Re-estimation and rebalancing dates must be specified in advance; otherwise information leakage can make an impossible strategy look successful.

Compare at least three models on the same universe and constraints: the SIM growth covariance, the full sample growth covariance, and a simple baseline such as equal weighting or a broad index. Record expected growth, modeled variance, realized wealth change and drawdown, turnover, costs, and constraint activity. A lower realized variance in one test window is evidence about that window, not proof of universal dominance.

Parameter uncertainty must also reach the portfolio decision. Refit the SIM over bootstrap or rolling samples, preserve each asset's joint $(\alpha, \beta, \sigma_{g,\varepsilon})$ draw, re-solve the optimizer, and inspect the distributions of weights, risk estimates, and variance regret. Comparing empirical-residual and Gaussian parametric propagation exposes sensitivity to the innovation assumption. If small input changes produce radically different positions, constraints, shrinkage, or a simpler portfolio may be more defensible than the nominal optimum.

KEY TAKEAWAYS

In this lecture, we inserted the SIM's low-rank-plus-diagonal moments into constrained risky-only and risky/risk-free allocation problems and separated model geometry from implementable trading rules.

- **Covariance validity is not forecast validity.** The SIM matrix is automatically positive semidefinite, but omitted residual dependence and unstable parameters can still make it a poor future-risk estimate.
- **Risk-free constraints change the frontier.** Lending, borrowing, short sales, and position bounds determine whether the CAL is a segment, a ray, or only part of a kinked efficient boundary.
- **Validation follows the decision clock.** Fit on past data, freeze the portfolio, evaluate later outcomes with costs, and test whether parameter uncertainty destabilizes weights.

Next, allow portfolio weights to evolve through time and distinguish a rebalancing policy from a one-time static optimum.

ADDITIONAL READING

- William F. Sharpe, "A Simplified Model for Portfolio Analysis," *Management Science* 9(2), 277–293 (1963).
- Harry Markowitz, "Portfolio Selection," *Journal of Finance* 7(1), 77–91 (1952).
- U.S. Department of the Treasury, "STRIPS: Separate Trading of Registered Interest and Principal of Securities," for the mechanics of stripped Treasury cash flows.